AITHON 2.0

ADAPTED AI

Personalized Learning That Adapts to Every Student

Problem Statement Track — EdTech

Problem Statement Title — AdaptEd AI - Personalized Learning Assistant.

Theme — Smart Education.

PS Category — Software

Team Name —**SkillMatix.** **Team ID - TEAM-634**

Team Members –

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IDEA TITLE

PROJECT:
AdaptED

ONE-LINE TAGLINE:
"Personalized Learning That Adapts to Every Student"

01 | THE REAL-WORLD PROBLEM



Every student has a different level, pace, goal and time. Classrooms and most digital tools give everyone the same material, same difficulty, same sequence.



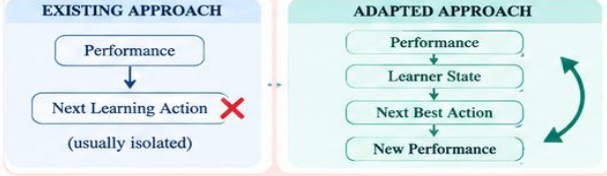
02 | WHY EXISTING TOOLS FALL SHORT

Generic AI chatbots
— answer a question, remember nothing about the learner

One-shot study planners
— plan once, never change

Static courses & videos
— same path for everyone, no gap detection

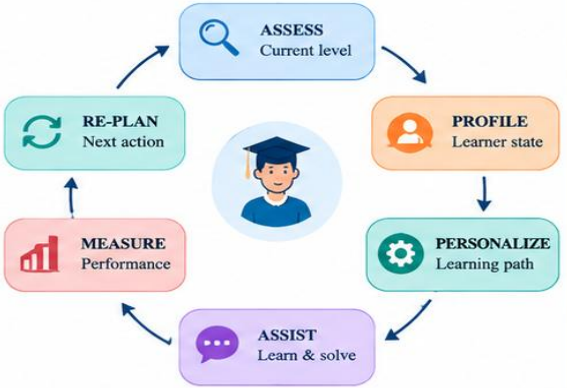
THE CORE GAP



03 | PROPOSED SOLUTION

AdaptED

Uses a dynamic learner model and continuous feedback to decide the next best learning action for every student.



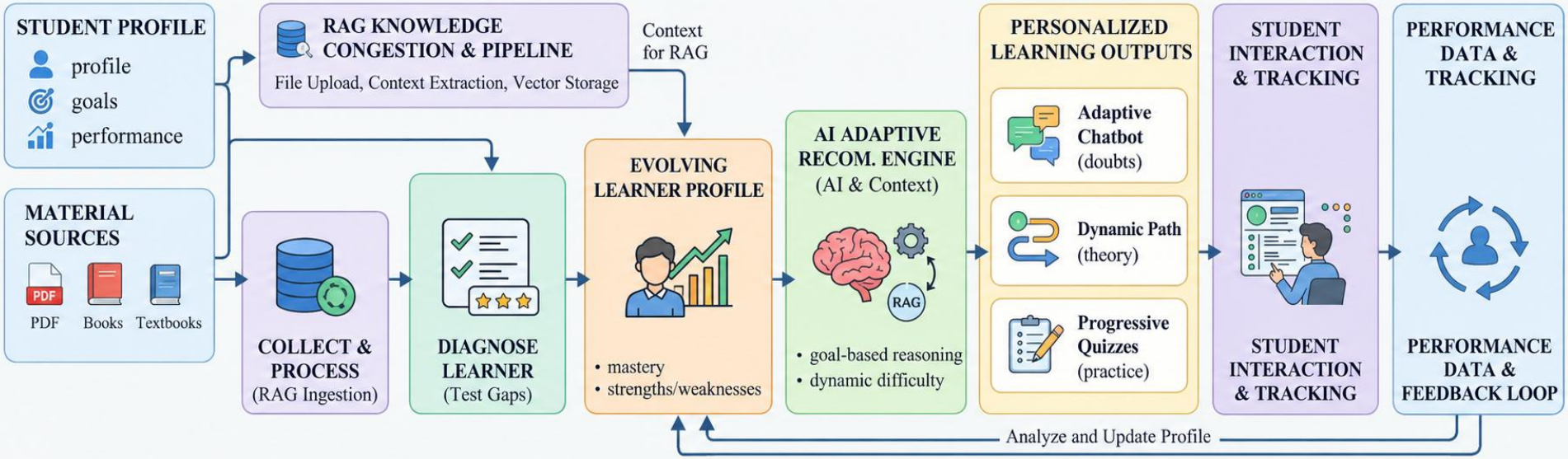
WHY IT NEEDS AN INTELLIGENT SOLUTION

UNDERSTAND learner state → **DETECT** hidden gaps → **DECIDE** next best step → **ADAPT** after every attempt

Same subject ≠ same learner ≠ same learning path

TECHNICAL APPROACH

Full-stack AI + RAG + Adaptive Learning Architecture



TECHNOLOGY STACK

FRONTEND
- Next.js
- React
- TypeScript
- React
- Material UI

BACKEND
- FastAPI
- Python
- Postgres / Supabase

AI & RAG
- Gemini API

RAG Tools:
- PyMuPDF
- Sentence Transformers
- ChromaDB

ANALYTICS & DEV
- Recharts
- Git
- GitHub

FEASIBILITY AND VIABILITY

Practical to build, scalable to deploy, and designed for real student learning needs.

Feasibility

Technical Feasibility

- Uses key mature technologies (e.g., Python, Gemini API, RAG). Modular architecture. (Tech list simplified)

Cost Feasibility

- Low MVP infrastructure needed.
- Uses open-source components and lightweight cloud architecture.

Implementation Feasibility

Frontend → API → RAG Model → Learner Model → Dashboard

Scalability

- Future Expansion: Multilingual support, extended subject and student base, institutional deployment, and mobile app.

Challenges & Risks

RAG Accuracy: Content validation required.

AI Hallucination: Quality assurance of outputs.

API Dependency: Latency & limits management.

Student Data Privacy: Protection of personal & performance data.

Limited Initial Data: Impact on initial modeling.

Mitigation Strategies

Step 1

RAG Validation: Better context grounding.

Step 2

Minimal Data Collection: Privacy and security design.

Step 3

Feedback & Evaluation: Quality control and refinement.

Viability

[Server] Low Infrastructure | [Puzzle] Modular | [Chart up] Scalable

IMPACT AND BENEFITS

*From one-size-fits-all learning to a continuously adapting **learning journey.***

Student → Learner Assessment → Dynamic Learner Model → Personalized Learning → Targeted Practice & Feedback → Continuous Adaptation

Outcome Panel
- Improved Learning Outcomes
- Engagement
- Retention
- Performance

01 | USER BENEFITS

Personalized Learning
Learning content and next actions respond to the learner's goals, mastery and activity.

Targeted Practice
Weak concepts can be identified and used to generate focused practice.

Grounded AI Assistance
Questions can be answered using the learner's uploaded learning material through RAG.

Progress Visibility
Mastery, strengths, weak topics and study history are surfaced through the dashboard.

Parent Visibility
The parent portal provides access to learning progress and activity information.

02 | SOCIAL IMPACT

Personalized Access
Supports learners who progress at different speeds and have different goals.

Self-Learning Support
Provides an always-available AI learning assistant for independent study.

Regional-language Expansion
Future Marathi/Hindi support can make the platform more accessible to regional-language learners.

Parent–Student Visibility
Helps parents understand learning activity and progress.

03 | ECONOMIC IMPACT

Low-cost Digital Infrastructure
Reduces the need for expensive physical coaching and multiple tools.

Reduced Tool Dependency
Minimizes the need for multiple disconnected learning platforms.

Reusable Across Subjects
The same platform and framework can be applied to multiple subjects and grades.

04 | EFFICIENCY & PRODUCTIVITY

Less Time Searching
Reduces time spent searching for relevant content.

Automated Quiz / Summary Creation
Saves time for both learners and educators.

Continuous Mastery Planning
Enables adaptive, mastery-based study plans.

05 | ENVIRONMENTAL IMPACT

Digital-First Learning Material
Reduces use of printed study material.

Lower Carbon Footprint
Supports a more sustainable learning ecosystem.

IMPACT PATHWAY

PROBLEM
Same learning path for different learners

→

ADAPTED SOLUTION
Learner model + RAG + mastery + adaptive engine

→

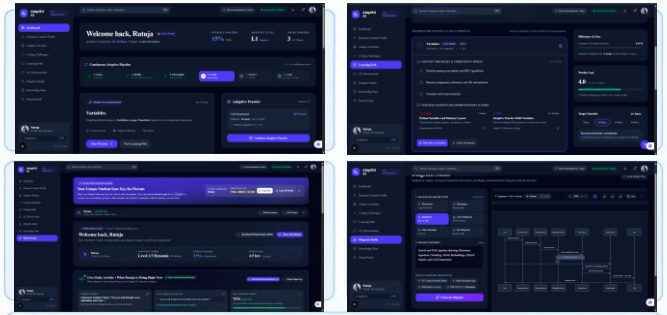
IMMEDIATE IMPACT
Better targeted practice, clearer progress visibility

→

LONG-TERM IMPACT
Continuous personalized learning

RESEARCH AND REFERENCES

01 WORKING PROTOTYPE



CURRENTLY FUNCTIONAL

RAG Chat • Quiz Engine • EMA Mastery • Dynamic Roadmap
Analytics • Parent Portal

02 12-HOUR FINALE ROADMAP

H0 – H2	H2 – H4	H4 – H7	H7 – H10	H10 – H12
SETUP & REFINEMENT	CORE ADAPTATION	FEATURE SPRINT	INTEGRATION & TESTING	DEMO & VALIDATION
• Environment setup • Prototype stabilization • Database/API check • Finalize demo flow	• Learner profile • Mastery state • Adaptive recommendations • RAG verification	• Logic Lab • Memory-Aware Engine • Adaptive practice integration	• End-to-end testing • UI refinement • Error handling • Performance checks	• Final test cases • Demo dataset • Presentation flow • Backup check

03 DEMO FLOW

1. **PROFILE** — Set goal & level
2. **UPLOAD** — Add learning material
3. **ASK / LEARN** — RAG-powered tutor
4. **TEST** — Quiz + mastery update
5. **ADAPT** — Next action / roadmap

04 FEATURE ROADMAP

CORE PROTOTYPE
- RAG Chat & AI Tutor
- Learner Profiling
- Quiz Engine
- EMA Mastery
- Dynamic Roadmap
- Analytics
- Parent Portal

SPRINT
- Logic Lab
- Memory-Aware Engine
- Marathi / Hindi Mode

STRETCH — IF TIME PERMITS
- Physical Gate Trainer (ESP32 → Breadboard → Truth-table verification)

05 PROTOTYPE STATUS

WORKING CORE → END-TO-END FLOW → DEMO READY → SPRINT → FINALE

06 RESOURCES

- Live Prototype →
- GitHub Repository →
- Documentation →
- Demo Materials →

PROTOTYPE STATUS | ✓ Working Core ✓ End-to-End Flow ✓ Demo Ready → Sprint Features (12-Hour Finale) → Stretch Feature (If Time Permits)